Lab 1: Requirements Engineering & UML Use-Case Modelling

PROBLEM STATEMENT #38 — Retail, E-Commerce & Finance

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Dropshipping Inventory & Order Orchestrator

1. Problem Context & Overview

An automated e-commerce backend that receives storefront order webhooks, validates incoming orders, checks third-party supplier inventory in real time, splits multi-vendor orders, automatically dispatches supplier purchase orders, handles out-of-stock items, and tracks shipment tracking numbers.

Target Stakeholders / Actors

- **Storefront Platform** — Sends customer order information to the system through webhooks.
- **E-Store Owner** — Monitors orders, handles exceptions, and updates tracking information.
- **Supplier Partner** — Provides supplier inventory information and receives purchase orders.

2. FUNCTIONAL REQUIREMENTS TABLE

ID & Priority	Requirement Description	Acceptance Criteria	Rationale
FR-001 — High	The system shall receive order webhooks from configured storefront platforms and register each valid order for processing.	**Pass:** A valid storefront webhook is received, assigned a unique order/event ID, and registered for processing. **Fail:** A valid webhook is received but the order is not registered.	Ensures customer orders are reliably captured from the storefront platform.
FR-002 — High	The system shall validate each incoming order webhook by checking the payload, required order fields, authentication/signature, and product information.	**Pass:** Valid order webhooks are accepted for processing and invalid requests are rejected. **Fail:** An invalid or unauthorized webhook is processed as a valid order.	Prevents malformed, incomplete, or unauthorized orders from entering the processing pipeline.
FR-003 — High	The system shall process multi-vendor orders by identifying the	**Pass:** Each order item is mapped to its supplier and	Ensures orders are correctly routed to

	supplier associated with each ordered product and checking supplier stock availability in real time.	current supplier inventory is verified. **Fail:** An order is dispatched without verifying supplier stock.	suppliers and prevents unnecessary purchase orders for unavailable products.
FR-004 — High	The system shall automatically split multi-vendor orders and generate supplier purchase orders for available products.	**Pass:** Products belonging to different suppliers are separated and the appropriate supplier purchase orders are generated. **Fail:** Products are incorrectly assigned to suppliers or required purchase orders are not generated.	Enables automated fulfillment of orders involving multiple suppliers.
FR-005 — High	The system shall identify out-of-stock supplier items and handle them according to the configured business policy without dispatching an invalid purchase order.	**Pass:** Out-of-stock items are identified and the configured exception handling process is triggered. **Fail:** A purchase order is sent for an unavailable supplier item.	Prevents failed fulfillment and allows the store owner to handle unavailable products appropriately.
FR-006 — Medium	The system shall allow the E-Store Owner to update shipment tracking numbers and maintain the latest tracking information for customer orders.	**Pass:** The owner can update a valid tracking number and the updated information is stored against the correct order. **Fail:** Tracking information is not updated or is associated with the wrong order.	Keeps shipment information accurate and enables customers and administrators to monitor deliveries.
FR-007 — Medium	The system shall allow the E-Store Owner to view order status, supplier fulfillment status, and shipment tracking information.	**Pass:** The owner can retrieve an order and view its current fulfillment and tracking status. **Fail:** Required order or tracking information is unavailable.	Provides visibility into the complete order fulfillment lifecycle.

3. NON-FUNCTIONAL REQUIREMENTS TABLE

ID & Priority	Requirement Description	Acceptance Criteria	Rationale

NFR-001 — High	The system shall acknowledge valid storefront webhook requests within **200 ms** under normal operating conditions to prevent webhook retry storms.	**Pass:** Benchmark testing confirms that at least 95% of valid webhook requests receive an HTTP 200 acknowledgement within 200 ms under the defined load. **Fail:** More than 5% of requests exceed the specified acknowledgement time.	Ensures fast webhook acknowledgement and prevents duplicate webhook deliveries caused by provider timeouts.
NFR-002 — High	The system shall authenticate storefront and supplier communication, use secure communication channels, and protect order and customer information from unauthorized access.	**Pass:** Unauthorized requests are rejected, communication uses secure protocols, and protected order information cannot be accessed without authorization. **Fail:** Unauthorized requests are accepted or sensitive order information is transmitted insecurely.	Protects customer, order, and supplier information and ensures secure communication between external systems.
NFR-003 — High	The system shall ensure that repeated delivery of the same storefront webhook or order event does not result in duplicate purchase orders or duplicate business operations.	**Pass:** Repeated delivery of the same unique order/event ID results in only one effective order processing operation. **Fail:** Duplicate webhook deliveries generate duplicate supplier purchase orders.	Prevents duplicate supplier orders and financial losses caused by webhook retries.
NFR-004 — Medium	The system shall maintain order processing logs, supplier stock checks, purchase order details, out-of-stock events, tracking updates, timestamps, and status changes for auditing and troubleshooting.	**Pass:** Administrators can retrieve the required order, supplier, purchase order, tracking, and processing information for the configured retention period. **Fail:** Required processing or order history information is missing.	Provides traceability and simplifies troubleshooting of fulfillment failures and customer-order issues.

4. USE CASE FLOW SPECIFICATION

Use Case ID: UC-001

Use Case Name: Process Multi-Vendor Order

Primary Actor: Storefront Platform

Secondary Actors: E-Store Owner, Supplier Partner, Dropshipping Inventory & Order Orchestrator

Goal

The storefront platform wants to send a customer order to the system so that the order can be validated, supplier stock can be checked, products can be divided among multiple suppliers, purchase orders can be dispatched, and shipment tracking can be maintained.

Preconditions

1. The storefront platform is registered and configured in the system.
2. The webhook endpoint is available.
3. The storefront webhook contains the required order information.
4. Supplier partners and their product inventory information are configured.
5. Supplier APIs are available for inventory verification.
6. The order contains valid product and customer information.

Postconditions

1. The customer order is stored with a unique order ID.
2. The order is validated successfully.
3. Supplier stock availability is determined for each order item.
4. Multi-vendor items are separated according to their suppliers.
5. Supplier purchase orders are generated for available products.
6. Out-of-stock products are handled according to the configured policy.
7. Supplier and order fulfillment status is recorded.
8. Tracking information can be updated and associated with the order.

Main Success Scenario

1. The **Storefront Platform** sends an order webhook to the system.
2. The system receives the incoming order webhook.
3. The system generates or identifies a unique order/event ID.
4. The system validates the webhook authentication/signature and required order fields.
5. The system stores the valid order.
6. The system identifies all products contained in the order.
7. The system identifies the supplier associated with each product.
8. The system sends inventory availability requests to the appropriate suppliers.
9. The system receives the current supplier stock information.
10. The system verifies that the required products are available.

11. The system groups order items according to their respective suppliers.
12. The system creates separate supplier purchase orders for the different suppliers.
13. The system dispatches the purchase orders to the appropriate supplier partners.
14. The system records the supplier purchase order information.
15. The system updates the order fulfillment status.
16. The supplier processes the purchase order and provides shipment information.
17. The E-Store Owner updates the tracking number when available.
18. The system stores the tracking number against the corresponding order.
19. The E-Store Owner can view the order and current tracking status.

5. ALTERNATE FLOWS

AF-01: Invalid Order Webhook

At **Step 4**, if the webhook authentication, signature, payload, or required order information is invalid:

1. The system detects that the webhook is invalid.
2. The system rejects the webhook request.
3. The system records the validation failure.
4. The system returns an appropriate error response to the storefront platform.
5. The order is not processed.
6. No supplier purchase order is generated.

AF-02: Supplier Product Out of Stock

At **Step 10**, if one or more ordered products are unavailable from the supplier:

1. The system identifies the unavailable product.
2. The system records the out-of-stock condition.
3. The system prevents a purchase order from being generated for the unavailable item.
4. The system applies the configured out-of-stock handling policy.
5. The system notifies or provides the exception information to the E-Store Owner.
6. Available products may continue through supplier purchase-order processing if permitted by the configured policy.
7. The order status is updated to reflect the out-of-stock condition.

AF-03: Multiple Suppliers in One Order

At **Step 11**, if the customer order contains products from multiple suppliers:

1. The system identifies the supplier for each order item.

2. The system groups products according to supplier.
3. The system creates a separate purchase order for each supplier.
4. The system sends each purchase order to its respective supplier.
5. The system maintains a relationship between the original customer order and all supplier purchase orders.
6. The system tracks the fulfillment status of each supplier separately.

AF-04: Supplier API Failure

At **Step 8**, if a supplier inventory API is unavailable or returns an error:

1. The system records the supplier API failure.
2. The system does not assume that the product is available.
3. The system marks the supplier inventory check as failed or pending.
4. The system retries the supplier inventory request according to the configured retry policy.
5. If the retry succeeds, the system continues order processing.
6. If the retry limit is reached, the system records the failure and alerts the E-Store Owner or marks the order for manual handling.

AF-05: Duplicate Order Webhook

At **Step 3**, if the system detects that the order/event ID has already been processed:

1. The system identifies the webhook as a duplicate.
2. The system does not create another customer order.
3. The system does not generate another supplier purchase order.
4. The system records the duplicate delivery attempt.
5. The system returns an appropriate acknowledgement response to the storefront platform.

AF-06: Tracking Number Update

After supplier fulfillment, when the supplier provides shipment information:

1. The E-Store Owner receives the shipment/tracking information.
2. The owner enters or updates the tracking number in the system.
3. The system validates the order reference and tracking information.
4. The system associates the tracking number with the correct order.
5. The system updates the shipment status.
6. The updated tracking information becomes available when viewing the order.

6. USE CASES IDENTIFIED FROM THE UML DIAGRAM

The system contains the following major use cases:

1. **UC-01 — Receive Order Webhook**
2. **UC-02 — Validate Order Webhook**
3. **UC-03 — Process Multi-Vendor Order**
4. **UC-04 — Check Supplier Stock**
5. **UC-05 — Split/Dispatch Purchase Orders**
6. **UC-06 — Handle Out-of-Stock Item**
7. **UC-07 — Update Tracking Number**
8. **UC-08 — View Order & Tracking Status**

Actor Responsibilities

Storefront Platform

- Sends order webhooks.
- Provides customer order information.

E-Store Owner

- Handles out-of-stock/exception cases.
- Updates tracking numbers.
- Views order and tracking status.

Supplier Partner

- Provides current stock information.
- Receives supplier purchase orders.
- Provides fulfillment/shipment information.

Main Use-Case Relationships

- **Receive Order Webhook** includes **Validate Order Webhook**.
- **Validate Order Webhook** includes **Process Multi-Vendor Order**.
- **Process Multi-Vendor Order** includes **Check Supplier Stock**.
- **Process Multi-Vendor Order** includes **Split/Dispatch Purchase Orders**.
- **Split/Dispatch Purchase Orders** is extended by **Handle Out-of-Stock Item** when a required product is unavailable.
- **E-Store Owner** can perform **Update Tracking Number**.
- **E-Store Owner** and **Supplier Partner** can participate in order/tracking status visibility according to their system permissions.

7. SUMMARY

The **Dropshipping Inventory & Order Orchestrator** automates the complete flow from receiving an e-commerce order webhook to supplier fulfillment and shipment tracking. The system validates incoming orders, checks real-time supplier inventory, handles multi-vendor orders by splitting them into supplier-specific purchase orders, manages out-of-stock products, and maintains tracking information.

The functional and non-functional requirements ensure that the system is **fast, secure, reliable, duplicate-safe, auditable, and suitable for automated e-commerce order fulfillment**.

